

ASSESSMENT GUIDE

Unit: ITEC610 Python fundamentals for data science, Semester 2, 2024

**Assessment Artefact: Report, Python Code
and Presentation**

Assessment number (3)

Weighting [40%]

Why this assessment?

- The purpose is to assess students' comprehensive Python data science skills and understanding from data processing to data visualization on real-world datasets with consideration of data ethics.

What are the types of employability skills that I will acquire upon completion of this assessment?

Skill Type	
Developed critical and analytical thinking	☒
Developed ability to solve complex problems	☒
Developed ability to work effectively with others	☐
Developed confidence to learn independently	☒
Developed written communication skills	☒
Developed spoken communication skills	☐
Developed knowledge in the field study	☐
Developed work-related knowledge and skills	☒

Assessment Overview:

Purpose, as written in the EUO

Due date:	25 October 2024 - 5:00 PM
Weighting:	40% - (15 % Viva / Presentation, 25% Report)
Length and/or format:	Individual Python source code with detailed comments and discussion within Colab notebook. The project report analyses the output and figures of the source code following structures in the 'Structure' section below. In-class Viva / Presentation.
Learning outcomes assessed	LO3, LO4
Graduate attributes assessed	GA3, GA4, GA5
How to submit:	via LEO
Return of assignment:	via LEO within 2 weeks of submission
Assessment criteria:	Rubric: see end of document

Context

Data processing, analysis and visualisation project

In this project, you will be given a dataset for data processing, analysis and visualisation.

We will be analysing the BRFSS (brfss.csv) weight vs height data. The Behavioral Risk Factor Surveillance System (BRFSS) is the premier system of health-related telephone surveys that collect state data about U.S. residents regarding their health-related risk behaviours, chronic health conditions, and use of preventive services. Weight and height are queried in a telephone interview, and the data for this project can be downloaded from the unit CANVAS website, and Pandas can be used to load it.

The six columns in the data represent: age, current_weight (kg), weight_a_year_ago (kg), current_weight with two decimals(kg), height (cm), and gender, where gender == 1 represents male and 2 represents female.

In this project, you are required to make insightful discoveries about the data via initial exploratory and visualisation with the skills learned from this unit.

Instructions

Attempt below tasks with the given dataset, at the same time, reflect on the development and applications of data science while ensuring the respect of human rights and of the values shaping open, pluralistic and tolerant information societies.

Task 1 (10 marks): Produce a **summary statistics** graph on current_weight, weight_a_year_ago, and height. [Hint: similar to Figure 1 below]

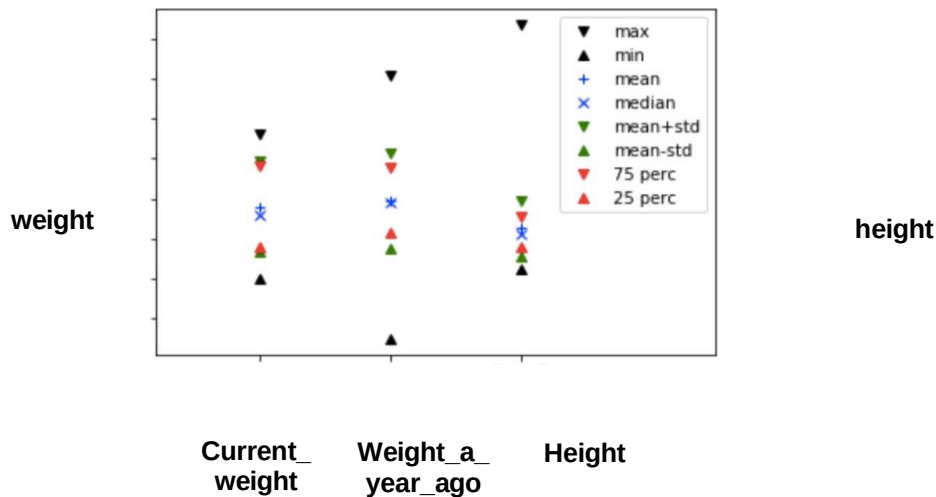


Figure 1: An example of summary statistics graph

Task 2 (9 marks): Calculate correlation: Define $\text{weight_change} = (\text{current_weight} - \text{weight_a_year_ago})$. Calculate the correlation between weight_change and the following variables, and determine which one is most correlated (regardless of the sign of correlation) with weight_change. Use **scatter plots** to support your conclusion.

- current_weight
- weight_a_year_ago
- age

[Hint: One scatter plot for each variable.]

Task 3 (6 marks)

Given a small dataset of house sizes and their respective prices, use scikit-learn to create a simple linear regression

model and predict a house price of 2500 sqft.

```
house_sizes = [[1500], [2000], [2500], [3000], [3500]]
```

```
house_prices = [250000, 300000, 330000, 360000, 390000]
```

Task 4: Write a report summarising insightful discoveries about the output and figures of the source code. The report should contain at least 3 sections; each should address a discussion of one of the above tasks. An extra conclusion/summary section could also be included. Below 'Structure' section gives an example of the report skeleton you could follow.

Structure

Prepare a Colab Notebook for this project. The structure of the Colab Notebook should alternate texts and python codes and cover topics listed the in specific tasks above. One template could be found in any week's workshop resources in LEO.

Prepare a report with the below skeleton as an example:

Title: Exploring BRFSS data

A brief introduction about the dataset, e.g., background of the data (search from web) and how many rows and columns it has.

Section 1: summary statistics analysis

In this section, describe the statistics you used in your code and then attach the figure you've obtained from the visualization. Then, do a comparison of the statistics for the three variables.

Section 2: correlations analysis

In this section, describe the formula you used to calculate the correlation in your code and present the numbers and scatter plots you obtain for the correlations between weight change and the three factors mentioned in task 2. Then do a quick comparison for the three factors.

Section 3: Significant analysis

In this section, summarize the results you obtain for Task 3.

Section 4: Conclusion

Sum up your overall insights about this dataset.

Naming the submission: start with your student ID, name and followed by unit name and code, i.e., studentID_studentNames ITEC610_AT3.ipynb, e.g., S00258769_Alice_Zuk ITEC610_AT3.ipynb/ S00258769_Alice_Zuk ITEC610_AT3.pdf. (.ipynb for code and .pdf for report)

How do I submit?

Submit Colab Notebook (.ipynb), the pdf version of ipynb file, and report (.pdf) to Assessment 3 via CANVAS assessment tile.

Note that Both the code and the report will be compared to other students' submissions in Turnitin to make sure the submission satisfies academic integrity.

Some Helpful Websites and Resources

Anaconda environment <https://docs.anaconda.com/anaconda/>

Python official website <https://www.python.org/>

Useful python packages:

<https://numpy.org/>
<https://pandas.pydata.org/>
<https://matplotlib.org/>

Who can help me?

Studiosity

Academic skills Unit (ASU)

Dr Walayat Hussain

I'm having problems

Application for Extension (EX) of Time for submission of an Assessment Task: The [EX form](#) should be completed by ACU students applying for an extension of time for submission of an assessment task. The completed and signed form must be submitted to the relevant National Lecturer-in-Charge prior to the due date of the assessment task. It must be accompanied by supporting documentary evidence such as an EIP, a doctor's certificate or equivalent, a death certificate, or a statutory declaration.

Special Consideration: Students use this form to apply for Special Consideration for assessable work in their studies at Australian Catholic University. Approval of such applications will only be granted to students who are legitimately disadvantaged in their assessment due to exceptional and unforeseen **circumstances** beyond their control.

Referencing

All referencing should be in [ACU Harvard style](#); however if you are coming from another faculty, you may choose to use your usual referencing style. If this is the case you must indicate at the top of your reference list what referencing style you are using (e.g. APA, MLA, Chicago, etc).

Please ensure your assignment makes use of in-text citations and a reference list. Missing citations or references is equivalent to plagiarism.

Criteria

The full criteria is compiled in a rubric, which can be found on the following page/s.

Rubric for Assessment 3

Relevant LO/GAs	Criterion (related to a single GA from the related LO – one GA per criterion)	Does not meet expectations	Meets expectations	Exceeds expectations		
		NN (0-49%)	PA (50-64%)	CR (65-74%)	DI (75-84%)	HD (85-100%)
GA5 LO3 and LO4 Weight=20 marks TL=3 Learning stage = I and D	Demonstrate correct understanding of the concepts of data processing, analysis and visualisation. Through demonstrating in class and in report.	Fail to adequately demonstrate correct understanding of the concepts of data processing, analysis and visualisation, i.e., None of the above tasks are addressed and no figures are produced.	Adequately demonstrate correct understanding of the concepts of data processing, analysis and visualisation, i.e., at least one task is addressed and one figure is produced with reasonable quality	Credibly demonstrate correct understanding of the concepts of data processing, analysis and visualisation, i.e., at least two tasks are addressed and one figure is produced with desired quality .	Distinctively demonstrate correct understanding of the concepts of data processing, analysis and visualisation, i.e., most of the tasks are addressed and the figures are produced with desired quality .	Highly distinctively demonstrate correct understanding of the concepts of data processing, analysis and visualisation, i.e., all tasks are addressed with figures of desired quality .
GA4 LO3 Weight=15 marks TL=3 Learning stage = I and D	Demonstrate critical and reflective thinking skills by observing and summarizing output of codes and figures	Fail to adequately demonstrate critical and reflective thinking skills by observing and summarizing output of codes and figures, i.e., no summary and conclusion are drawn around the output in the report	Adequately demonstrate critical and reflective thinking skills by observing and summarizing output of codes and figures, i.e., thoughts about the output of the completed tasks are drawn in the report	Credibly demonstrate critical and reflective thinking skills by observing and summarizing output of codes and figures, i.e., reasonable insights about the output of the completed tasks and figures are given in the report	Distinctively demonstrate critical and reflective thinking skills by observing and summarizing output of codes and figures, i.e., thoughts and insights about the code output and the figures of the completed tasks are detailed in the report	Highly distinctively demonstrate critical and reflective thinking skills by observing and summarizing output of codes and figures, i.e., thoughts and insights about all the tasks and figures are detailed in the report
GA3 LO4 Weight=5 marks TL=3	Demonstrate awareness of data ethics occurred in the data science process	Fail to adequately demonstrate awareness of data ethics occurred in the data science process, i.e., no data is explored	Adequately demonstrate awareness of data ethics occurred in the data science process, i.e., part of the data is explored via one task	Credibly demonstrate aware of data ethics occurred in the data science process, i.e., part of the data is explored via two tasks	Distinctively demonstrate awareness of data ethics occurred in the data science process, i.e., part of the data is explored via most of the tasks	Highly distinctively demonstrate awareness of data ethics occurred in the data science process, i.e., full data is explored via all tasks.